

M08 PROTECTED PRACTICE REVIEW

Open only after a genuine saved attempt

Compare concept/units/interpretation -> repair -> close -> changed retry.

STAT1 review - Descriptive statistics: center, spread and outliers

1. Strong attempt should demonstrate

Recompute one statistic independently, verify the row count, and explain whether any flagged outlier is a data error, rare valid case, or unknown.

2. Expected reasoning / worked pattern

Mean uses every value and is sensitive to extremes. Median is resistant to extreme observations. Standard deviation describes spread around the mean; IQR describes the middle 50%. An outlier flag is a reason to investigate, not a deletion command. Example: Delivery minutes [20, 22, 24, 25, 150]. The mean is pulled upward by 150 while the median stays near the typical routine delivery. For a highly right-skewed metric, report median plus percentiles/IQR alongside the mean rather than claiming one number is the whole story.

3. What to compare

Compare population/sample/metric, units, assumptions, calculation and interpretation before exact wording. Equivalent arithmetic is allowed when meaning and validation are correct.

4. Likely repair

Reporting only the mean for skewed data; automatically deleting an extreme value because it is inconvenient.

5. Fresh retry

Close this review. Complete the changed task from the matching canonical workbook sheet, save independent validation and append evidence. Only the fresh retry upgrades mastery.

STAT2 review - Distributions, percentiles, normal models and z-scores

1. Strong attempt should demonstrate

Check shape visually and compare mean/median before using normal-model language.

2. Expected reasoning / worked pattern

A distribution shows how values are spread. Percentiles express relative position. A z-score is $(\text{value} - \text{mean}) / \text{standard deviation}$. Normal models are useful only when the shape/assumptions are a reasonable approximation. Example: If mean handle time is 12 minutes and SD is 3, an 18-minute call has $z = +2$. It is two standard deviations above the mean. That makes it unusual relative to the model, not automatically invalid.

3. What to compare

Compare population/sample/metric, units, assumptions, calculation and interpretation before exact wording. Equivalent arithmetic is allowed when meaning and validation are correct.

4. Likely repair

Assuming every business metric is normal; calling a high z-score automatically wrong.

5. Fresh retry

Close this review. Complete the changed task from the matching canonical workbook sheet, save independent validation and append evidence. Only the fresh retry upgrades mastery.

STAT3 review - Population, samples and sampling bias

1. Strong attempt should demonstrate

State who could never be selected under your design and whether that limits the target population.

2. Expected reasoning / worked pattern

Random sampling error is chance variation between samples. Sampling bias is systematic and does not disappear just because the sample is large. Selection, nonresponse and survivorship can distort conclusions. Example: A survey sent only to recent support users can overrepresent frustrated customers. Ten thousand such responses can be less representative than a smaller probability sample of all active customers.

3. What to compare

Compare population/sample/metric, units, assumptions, calculation and interpretation before exact wording. Equivalent arithmetic is allowed when meaning and validation are correct.

4. Likely repair

Generalizing beyond the sampled population; believing sample size repairs selection bias.

5. Fresh retry

Close this review. Complete the changed task from the matching canonical workbook sheet, save independent validation and append evidence. Only the fresh retry upgrades mastery.

STAT4 review - Sampling distributions, standard error and the CLT

1. Strong attempt should demonstrate

Check that the observed spread of repeated sample means decreases with n while the underlying population spread stays similar.

2. Expected reasoning / worked pattern

A statistic such as a sample mean varies across repeated samples. Standard error describes variability of an estimator. For a mean, SE is commonly approximated by sample SD / $\sqrt{n}$. The CLT explains why sample means often become approximately normal under broad conditions as n grows. Example: If two surveys have the same SD but one has 9 times as many observations, its standard error is roughly one-third as large because $\sqrt{9}=3$. More precision does not repair a biased sample.

3. What to compare

Compare population/sample/metric, units, assumptions, calculation and interpretation before exact wording. Equivalent arithmetic is allowed when meaning and validation are correct.

4. Likely repair

Confusing SD with SE; confusing precision with representativeness.

5. Fresh retry

Close this review. Complete the changed task from the matching canonical workbook sheet, save independent validation and append evidence. Only the fresh retry upgrades mastery.

STAT5 review - Confidence intervals and uncertainty

1. Strong attempt should demonstrate

Verify the SE arithmetic and state at least one assumption/limitation.

2. Expected reasoning / worked pattern

A 95% confidence procedure is designed to capture the true parameter in about 95% of repeated samples under its assumptions. A narrower interval means greater precision, not automatically less bias. Always report the estimate itself. Example: Estimated conversion lift +0.5 percentage points with 95% CI [-0.2pp, +1.2pp] does not exclude zero at the 95% level. Positive business-relevant effects may still be plausible.

3. What to compare

Compare population/sample/metric, units, assumptions, calculation and interpretation before exact wording. Equivalent arithmetic is allowed when meaning and validation are correct.

4. Likely repair

Saying a completed CI has a 95% probability of containing a fixed true parameter; treating a narrow biased interval as trustworthy.

5. Fresh retry

Close this review. Complete the changed task from the matching canonical workbook sheet, save independent validation and append evidence. Only the fresh retry upgrades mastery.

STAT6 review - Hypothesis testing, null hypothesis and p-values

1. Strong attempt should demonstrate

State the null, alpha, estimate, p-value and decision separately.

2. Expected reasoning / worked pattern

The null hypothesis is the baseline/no-effect model. A p-value asks how surprising the observed data or more extreme data would be if the null were true. $p < \alpha$ is statistically significant under that rule; failing to reject H_0 is not proving H_0 . Example: If $p=0.08$ at $\alpha=0.05$, write: the data do not provide enough evidence to reject the null at 0.05. Do not write: there is no effect.

3. What to compare

Compare population/sample/metric, units, assumptions, calculation and interpretation before exact wording. Equivalent arithmetic is allowed when meaning and validation are correct.

4. Likely repair

Saying $p=0.03$ means a 3% probability the null is true; saying $p>0.05$ proves no effect.

5. Fresh retry

Close this review. Complete the changed task from the matching canonical workbook sheet, save independent validation and append evidence. Only the fresh retry upgrades mastery.

STAT7 review - Effect size, practical significance and decision thresholds

1. Strong attempt should demonstrate

State estimate, uncertainty, business threshold and guardrail separately.

2. Expected reasoning / worked pattern

Absolute lift = treatment rate - control rate. Relative lift = absolute lift / control rate. Tiny effects can be statistically significant in huge samples; valuable effects can be uncertain in small samples. Define a minimum worthwhile effect and guardrails before seeing results when possible. Example: Control conversion 5.0%, treatment 5.4%: absolute lift +0.4 percentage points; relative lift 8%. Whether +0.4pp matters depends on economics, risk and costs.

3. What to compare

Compare population/sample/metric, units, assumptions, calculation and interpretation before exact wording. Equivalent arithmetic is allowed when meaning and validation are correct.

4. Likely repair

Calling every statistically significant result important; ignoring a damaged guardrail metric.

5. Fresh retry

Close this review. Complete the changed task from the matching canonical workbook sheet, save independent validation and append evidence. Only the fresh retry upgrades mastery.

STAT8 review - Correlation, confounding and causal claims

1. Strong attempt should demonstrate

Inspect the scatter pattern and confirm that one or two extreme points are not driving the whole association.

2. Expected reasoning / worked pattern

Pearson correlation ranges from -1 to +1 for linear association. High correlation can be caused by confounding, reverse causality, selection effects or a real direct effect. Randomized experiments are usually stronger for causal inference. Example: Advertising spend and sales can be correlated because larger stores both spend more and sell more. The association can be real while the causal story is wrong.

3. What to compare

Compare population/sample/metric, units, assumptions, calculation and interpretation before exact wording. Equivalent arithmetic is allowed when meaning and validation are correct.

4. Likely repair

Calling correlation causal; ignoring common causes and reverse causality.

5. Fresh retry

Close this review. Complete the changed task from the matching canonical workbook sheet, save independent validation and append evidence. Only the fresh retry upgrades mastery.

STAT9 review - A/B tests and experimental thinking

1. Strong attempt should demonstrate

Confirm group counts, metric definitions, missing/duplicate IDs and guardrail integrity before interpreting the result.

2. Expected reasoning / worked pattern

Define treatment/control and unit of randomization. Predefine primary metric, guardrails, duration/stopping rule and decision threshold. Randomization helps balance confounders on average. Check sample-ratio problems, data loss and implementation failures. Avoid opportunistic peeking and post-hoc metric switching. Example: A clean question: among eligible visitors during a planned 14-day period, does a simplified signup page increase signup conversion without materially increasing support contacts?

3. What to compare

Compare population/sample/metric, units, assumptions, calculation and interpretation before exact wording. Equivalent arithmetic is allowed when meaning and validation are correct.

4. Likely repair

Changing metrics after results; repeatedly peeking and stopping at $p < 0.05$; presenting exploratory segments as confirmed effects.

5. Fresh retry

Close this review. Complete the changed task from the matching canonical workbook sheet, save independent validation and append evidence. Only the fresh retry upgrades mastery.

M-STAT project review - decision discipline

1. Protected objective controls

Control n=700, rate=0.1000; Treatment n=700, rate=0.1171; absolute lift=1.7143pp; relative lift=17.1429%; z=1.0309; p=0.3026; 95% CI=[-1.5438,4.9723]pp; support rates 6.00% vs 6.43%.

2. Interpretation boundary

The point estimate is positive but uncertainty includes zero and the two-sided p-value is above 0.05. Do not write 'no effect'. State the estimate/compatible effects and business/guardrail context separately.

3. If it failed

Descriptive/distribution -> STAT1/2; sample/design -> STAT3/9; SE/CI -> STAT4/5; p-value -> STAT6; effect/business threshold -> STAT7; causal claim -> STAT8.

4. Fresh retry

Close this review. Repair the weak concept on a changed calculation/example. Gate A remains a fresh assessment and does not open until the project evidence is complete.